CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 5: **Measurement of Length and Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A rope is 3.2 m long. After cutting off pieces of 45 cm, 55 cm and 60 cm, the remaining length is: 1

(a) 100 cm **(b)** 160 cm **(c)** 320 cm **(d)** 200 cm

2. In a pendulum experiment, 20 oscillations took 36 s. A second student gets 18 s for 10 oscillations. Both agree because: 1

(a) time period is 1.8 s in both **(b)** he is wrong
(c) 36 s vs 18 s differ **(d)** time period differs

3. A student measuring a pencil places the 2 cm mark of the scale at one end and reads 12.5 cm at the other end. Length is: 1

(a) 14.5 cm **(b)** 12.5 cm **(c)** 10.5 cm **(d)** 2.0 cm

4. A road is 2.7 km. A runner has run 1850 m. The remaining distance is: 1

(a) 1850 m **(b)** 750 m **(c)** 850 m **(d)** 650 m

5. Which of the following is true about a freely falling stone? 1

(a) it moves in a circle
(b) it is at rest
(c) it is in linear motion and its speed increases
(d) it is in periodic motion

6. If the actual length of an object is 8.0 cm and a student measures 8.3 cm, 8.2 cm and 8.4 cm, the error is likely: 1

(a) random only **(b)** systematic (consistently higher)
(c) zero **(d)** due to units

7. A wheel of circumference 2.5 m makes 480 turns. Distance moved is: 1

(a) 1200 m **(b)** 480 m **(c)** 120 m **(d)** 2400 m

8. Which of the following shows both rotational and linear motion at the same time? 1

(a) a swing **(b)** a fan **(c)** a pendulum **(d)** a rolling ball

9. Which of these is the best reason for taking repeated measurements? 1

(a) to reduce random errors **(b)** to make the object longer
(c) to waste time **(d)** to change units

10. A thread of length 188 cm just goes round a circular plate 4 times in a spiral winding of 4 turns. The circumference is: 1

(a) 47 cm **(b)** 94 cm **(c)** 752 cm **(d)** 188 cm

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A road runs 3.6 km. A boy walks at a step of 60 cm. How many steps does he take? 2

12. Convert 0.003 km into cm. 2

13. Why is the motion of a swing considered periodic but that of a falling leaf not? 2

14. Why can a circle's circumference not be measured with a straight ruler directly? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. The circumference of a wheel is 2.2 m. How many turns will it make to cover 1.1 km? 3

16. Describe an experiment to measure the length of a curved line of 1 metre length marked on a floor by two students. 3

17. A drawing is made to a scale of 1 cm for 5 m. A building is 40 m long. What is its length in the drawing? 3

OR

A thread wound 25 times round a pencil of uniform thickness measures 8 cm in length of the coil. Find the thickness of the thread. 3

18. Explain, with two examples each, the difference between rotation and revolution. 3

OR

Why do we say that a train passenger and platform observer disagree about speed? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. A student wants to find the circumference of a tree trunk, the thickness of a coin and the length of a pencil. Choose the instruments and explain the method for each. 5

OR

A student says 'My measurement of 12.4 cm is accurate because I measured only once'. Evaluate and suggest improvements. 5

20. A pendulum of length 1 m makes 30 oscillations in 60 s and a pendulum of length 0.25 m makes 30 oscillations in 30 s. Calculate their time periods and compare. 5

OR

Convert and compare: a snail moves 1.2 cm/s while a student walks 1.2 m/s. Express both in the same unit and find how many times the student is faster. 5

END OF QUESTION PAPER Total: 40 Marks